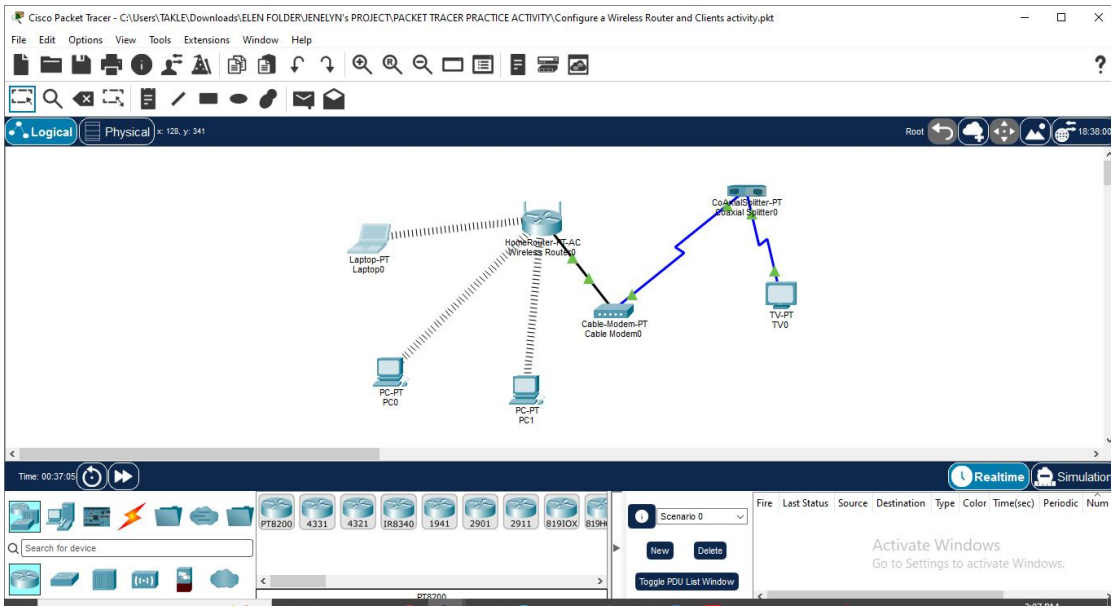
Project: Home Wireless Network Simulation — Cisco Packet Tracer

Objective:
Designed and tested a home network connecting a cable modem, wireless router, wired/wireless clients, and television service.

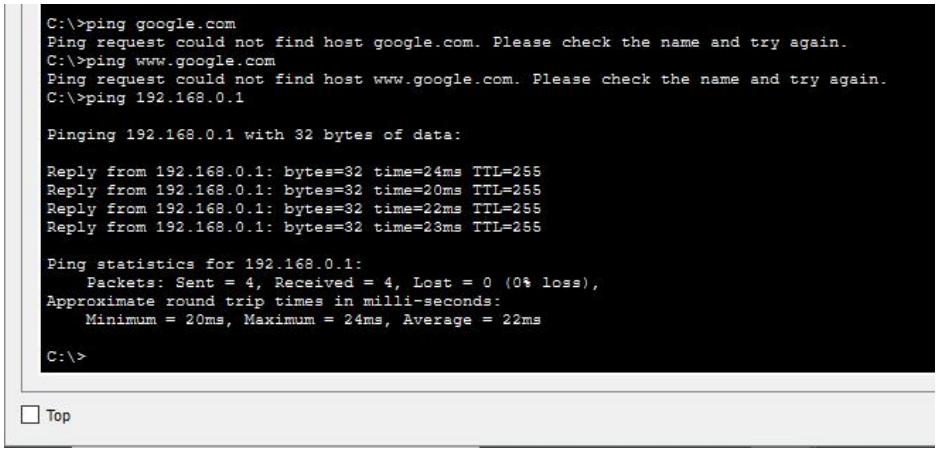


Technologies/Concepts demonstrated:

- IPv4 addressing
- DHCP
- Default gateway
- DNS
- Wi-Fi/SSID
- Wireless client connectivity
- LAN communication
- WAN/Internet connectivity
- ICMP/ping
- Coaxial networking
- Basic network troubleshooting

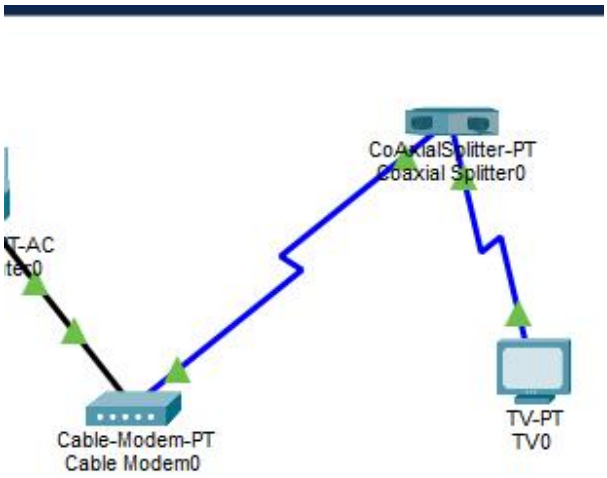
Testing performed:

✓ DHCP address assignment
✓ Default gateway connectivity
✓ Client-to-client connectivity
✓ Wireless connectivity
✓ Internet connectivity
✓ DNS resolution
✓ Failure/recovery testing



DNS Resolution Test — Troubleshooting Finding

The client was able to reach the configured DNS server by IP address, but google.com could not be resolved. The topology did not contain a dedicated DNS server, so external DNS resolution was not demonstrated in this simulation.



TV Connectivity Test

- ✧ TV connected is to the coaxial splitter
- ✧ Cable modem is connected to the splitter
- ✧ Link indicators are active

TV0 is connected to the coaxial splitter. The coaxial link between the splitter and TV0 is active. TV0 does not provide an IP/network configuration interface in this Packet Tracer model, so ICMP/DNS testing is not applicable.